

LAWSON EKHORUTOMWEN

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SUMMARY

Full-stack developer and AI engineer with backend depth — REST APIs, relational data modeling, and real-time systems — and a final-year medical student. Built and shipped production systems end to end: a broadcast pipeline processing 200,000+ messages/day, a four-app delivery platform on a shared Postgres-backed API, a multi-agent clinical documentation pipeline with built-in verification, and AI systems built from first principles rather than tutorials.

EXPERIENCE

Sendpiper — Frontend Engineer

Jul 2025 – Present

Meta-certified messaging automation platform for WhatsApp, Instagram & Messenger · Remote

- Built and maintain the customer-facing web application in React and Next.js, including a drag-and-drop flow builder and a unified inbox consolidating three separate messaging APIs (WhatsApp, Instagram, Messenger) into one real-time interface.
- Built the analytics and audience-management layer on top of a broadcast pipeline processing 200,000+ messages/day at 99.9% reliability.

PROJECTS

AI Clinical Scribe

Multi-agent pipeline turning a consult recording into a reviewed SOAP note: a LangGraph extraction agent is checked by separate QA and validator agents against the Whisper/pyannote.audio transcript before a doctor finalizes it. Three-pass QA measurably improved recall over a single pass in controlled testing. FastAPI, PostgreSQL with full edit history, React/TypeScript/Tailwind interface.

Rave — Food Delivery Platform

Designed and built the shared REST API (Node.js, Express, Prisma, PostgreSQL) powering four separate apps — customer, vendor, and rider (React Native/Expo), plus an internal admin dashboard (Next.js). Owned the data model and endpoints for auth, catalog, orders, and payments across all four consumers.

Ground — Full-Stack Medical RAG Assistant

Retrieval-augmented generation system answering health questions using only cited MedlinePlus sources. TF-IDF retrieval and cosine similarity built from scratch in PyTorch, with a hybrid fallback for corpus-common queries; FastAPI backend, Next.js/TypeScript frontend, deployed on Render and Netlify.

American Sign Language Translator

Real-time computer vision app translating ASL letters to text/speech and reversing text or voice into animated ASL sequences. MediaPipe hand-landmark extraction streamed over WebSocket to a PyTorch classifier; FastAPI backend, Next.js/React Three Fiber frontend, offline speech-to-text via Whisper.

Clinical NLP Pipeline

Built entity extraction and negation detection for medical text, implementing a NegEx-style algorithm from scratch. Evaluated against ground truth I hand-annotated myself, since no labeled dataset existed for the task.

GPT Language Model from Scratch

Implemented backpropagation, a custom autograd engine, attention, and a full transformer architecture without relying on high-level ML libraries to abstract away the core mechanics.

EDUCATION

Medicine and Surgery

University of Benin (UNIBEN) — Benin City, Edo State

Expected 02/2027

SKILLS

Backend & Data Node.js/Express, PostgreSQL, FastAPI, REST API Design, Prisma, SQLAlchemy, Docker

Full-Stack Development React, Next.js, Vue.js, React Native/Expo, Flask, TypeScript

AI/ML Engineering Neural Networks (Built from Scratch), Backpropagation & Autograd Engines, Transformer Architecture & Attention, LLM Training (GPT), LoRA Fine-Tuning, Retrieval-Augmented Generation (RAG), Multi-Agent Systems & LLM Orchestration (LangGraph, LangChain, Groq), Speech-to-Text & Speaker Diarization (Whisper, pyannote.audio), Explainable AI (SHAP, Grad-CAM, Integrated Gradients), Clinical NLP & Named Entity Recognition, Computer Vision (MediaPipe, OpenCV), MLOps, PyTorch

Languages Python, TypeScript / JavaScript

Infrastructure AWS, Google Cloud Platform (GCP), Render, Netlify